In [1]:
```
# Baseline OLS regressions: Table 2, Tables C.5 and C.6, Section 4.3
import os
import warnings
import numpy as np
import pandas as pd
import statsmodels.api as sm
from statsmodels.stats.stattools import durbin_watson

warnings.filterwarnings("ignore")
os.makedirs("output", exist_ok=True)
```

In [2]:
```
# Merge the clean files into one monthly data set, December 2015 to December 2025 (121
months)
files = ["spy_clean.csv", "rsp_clean.csv", "voo_clean.csv", "vix_clean.csv",
"dgs10_clean.csv",
         "AIStocks_ASVI8_Monthly.csv", "AIStocks_ASVI4_Monthly.csv",
         "mag7_clean.csv", "mag7b_clean.csv", "mag7c_clean.csv"]
df = pd.read_csv(files[0], parse_dates=["date"])
for f in files[1:]:
    df = df.merge(pd.read_csv(f, parse_dates=["date"]), on="date", how="inner")
df = df.sort_values("date").reset_index(drop=True)

# 2021-2025 subsample. December 2020 only serves as the first lag, so the VARs run on Jan
2021 to Dec 2025.
sub = df[df["date"] >= "2020-12-31"].reset_index(drop=True)
print(len(df), "months,", df["date"].min().date(), "to", df["date"].max().date())
```

```
121 months, 2015-12-31 to 2025-12-31
```

In [3]:
```
def stars(p):
    return "***" if p < 0.01 else "**" if p < 0.05 else "*" if p < 0.10 else ""


def ols(data, y, x):
    # Constant, regressor of interest and the two macro controls, HC3 robust standard errors
    X = sm.add_constant(data[[x, "dlog_vix", "d_dgs10"]])
    return sm.OLS(data[y], X).fit(cov_type="HC3")
```

In [4]:
```
# One-month lags. December 2015 only provides the lag, so the regressions run on Jan 2016 to
Dec 2025 (N = 120).
d = df.copy()
for c in ["ASVI8", "flow_spy", "flow_rsp", "flow_voo", "mag7_ret"]:
    d[c + "_lag"] = d[c].shift(1)
d = d.dropna().reset_index(drop=True)

# Sample windows for Tables C.5 and C.6
windows = {"2016-2025": "2016-01-31", "2020-2025": "2020-01-31", "2021-2025": "2021-01-31",
           "2022-2025": "2022-01-31", "2023-2025": "2023-01-31"}
window = {w: d[d["date"] >= start] for w, start in windows.items()}
print({w: len(data) for w, data in window.items()})
```

```
{'2016-2025': 120, '2020-2025': 72, '2021-2025': 60, '2022-2025': 48, '2023-2025': 36}
```

In [5]:
```
# Table 2: full sample
models = {
    "H1": ols(d, "flow_spy", "ASVI8_lag"),            # attention -> SPY flows
    "H1b": ols(d, "mag7_ret", "flow_spy_lag"),        # SPY flows -> Mag7 returns
    "H2": ols(d, "mag7_ret", "ASVI8_lag"),            # attention -> Mag7 returns
    "Feedback": ols(d, "ASVI8", "mag7_ret_lag"),      # Mag7 returns -> attention
}
for name, m in models.items():
    print(name)
    print(m.summary().tables[1], "\n")

rows = {"ASVI8_lag": "ASVI8 lag(1)", "flow_spy_lag": "Flow SPY lag(1)", "mag7_ret_lag": "Mag7
returns lag(1)",
        "dlog_vix": "dlogVIX", "d_dgs10": "dDGS10"}
table2 = {}
for name, m in models.items():
    col = []
    for var in rows:
```

```
        if var in m.params:
            col += [f"{m.params[var]:.3f}{stars(m.pvalues[var])}", f"({m.bse[var]:.3f})"]
        else:
            col += ["", ""]
    col += [int(m.nobs), f"{m.rsquared:.3f}", f"{durbin_watson(m.resid):.3f}", f"
{m.fvalue:.4g}"]
    table2[name] = col
index = [x for v in rows.values() for x in (v, "")] + ["N", "R2", "Durbin-Watson", "F-
statistic"]
table2 = pd.DataFrame(table2, index=index)
table2.to_csv("output/Table_2_OLS_full_sample.csv")
table2
```

H1

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0017	0.003	0.649	0.517	-0.003	0.007
ASVI8_lag	-0.0053	0.004	-1.325	0.185	-0.013	0.003
dlog_vix	-0.0331	0.012	-2.858	0.004	-0.056	-0.010
d_dgs10	0.0033	0.011	0.294	0.769	-0.019	0.026

H1b

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0234	0.006	3.901	0.000	0.012	0.035
flow_spy_lag	0.2708	0.218	1.244	0.214	-0.156	0.698
dlog_vix	-0.1572	0.027	-5.897	0.000	-0.209	-0.105
d_dgs10	-0.0463	0.028	-1.673	0.094	-0.101	0.008

H2

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0181	0.006	2.859	0.004	0.006	0.031
ASVI8_lag	0.0231	0.011	2.177	0.029	0.002	0.044
dlog_vix	-0.1521	0.026	-5.919	0.000	-0.202	-0.102
d_dgs10	-0.0411	0.027	-1.523	0.128	-0.094	0.012

Feedback

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.1808	0.050	3.640	0.000	0.083	0.278
mag7_ret_lag	2.3110	0.886	2.609	0.009	0.575	4.047
dlog_vix	-0.1673	0.210	-0.796	0.426	-0.579	0.245
d_dgs10	-0.1802	0.233	-0.774	0.439	-0.637	0.276

Out[5]:

	H1	H1b	H2	Feedback
ASVI8 lag(1)	-0.005		0.023**	
	(0.004)		(0.011)	
Flow SPY lag(1)		0.271		
		(0.218)		
Mag7 returns lag(1)				2.311***
				(0.886)
dlogVIX	-0.033***	-0.157***	-0.152***	-0.167
	(0.012)	(0.027)	(0.026)	(0.210)
dDGS10	0.003	-0.046*	-0.041	-0.180
	(0.011)	(0.028)	(0.027)	(0.233)
N	120	120	120	120
R2	0.095	0.267	0.287	0.092
Durbin-Watson	2.432	1.948	2.040	0.506
F-statistic	2.856	13.14	14.38	3.203

In [6]:

```
# Tables C.5 and C.6: all funds and sample windows
def cell(m, var, with_stars=False):
    return f"{m.params[var]:.4f}{stars(m.pvalues[var]) if with_stars else ''}
({m.bse[var]:.4f})"

funds = ["spy", "rsp", "voo"]
h1 = {(w, f): ols(window[w], f"flow_{f}", "ASVI8_lag") for w in windows for f in funds}
h1b = {(w, f): ols(window[w], "mag7_ret", f"flow_{f}_lag") for w in windows for f in funds}
h2 = {w: ols(window[w], "mag7_ret", "ASVI8_lag") for w in windows}
fb = {w: ols(window[w], "ASVI8", "mag7_ret_lag") for w in windows}

rowsC5 = []
for w in windows:
    rowsC5.append({"Panel": "A: Flow on ASVI8(t-1)", "Sample window": w, "N": len(window[w]),
                   **{f.upper(): cell(h1[(w, f)], "ASVI8_lag") for f in funds}})
for w in windows:
    rowsC5.append({"Panel": "B: Mag7 returns on Flow(t-1)", "Sample window": w, "N":
len(window[w]),
                   **{f.upper(): cell(h1b[(w, f)], f"flow_{f}_lag") for f in funds}})
tableC5 = pd.DataFrame(rowsC5)
tableC5.to_csv("output/Table_C5_OLS_flows_by_window.csv", index=False)

tableC6 = pd.DataFrame([{
    "Sample window": w, "N": len(window[w]),
    "H2: beta on ASVI8(t-1)": cell(h2[w], "ASVI8_lag", True), "H2 R2": f"
{h2[w].rsquared:.3f}",
    "H2 DW": f"{durbin_watson(h2[w].resid):.2f}",
    "Feedback: beta on Mag7(t-1)": cell(fb[w], "mag7_ret_lag", True), "Feedback R2": f"
{fb[w].rsquared:.3f}",
    "Feedback DW": f"{durbin_watson(fb[w].resid):.2f}",
} for w in windows])
tableC6.to_csv("output/Table_C6_OLS_returns_and_feedback.csv", index=False)

dw_a = [durbin_watson(m.resid) for m in h1.values()]
dw_b = [durbin_watson(m.resid) for m in h1b.values()]
print(f"Durbin-Watson: Panel A {min(dw_a):.2f} to {max(dw_a):.2f}, Panel B {min(dw_b):.2f} to
{max(dw_b):.2f}")
display(tableC5)
tableC6
```

```
Durbin-Watson: Panel A 1.37 to 2.72, Panel B 1.59 to 1.96
```

	Panel	Sample window	N	SPY	RSP	VOO
0	A: Flow on ASVI8(t-1)	2016-2025	120	-0.0053 (0.0040)	-0.0000 (0.0078)	0.0008 (0.0026)
1	A: Flow on ASVI8(t-1)	2020-2025	72	-0.0048 (0.0041)	-0.0014 (0.0094)	0.0013 (0.0032)
2	A: Flow on ASVI8(t-1)	2021-2025	60	-0.0032 (0.0041)	0.0022 (0.0102)	0.0009 (0.0029)
3	A: Flow on ASVI8(t-1)	2022-2025	48	-0.0028 (0.0048)	-0.0016 (0.0117)	-0.0004 (0.0029)
4	A: Flow on ASVI8(t-1)	2023-2025	36	-0.0049 (0.0057)	-0.0021 (0.0140)	-0.0032 (0.0025)
5	B: Mag7 returns on Flow(t-1)	2016-2025	120	0.2708 (0.2178)	-0.0163 (0.1768)	-0.0927 (0.3810)
6	B: Mag7 returns on Flow(t-1)	2020-2025	72	0.1766 (0.3164)	-0.1139 (0.2088)	0.0997 (0.5556)
7	B: Mag7 returns on Flow(t-1)	2021-2025	60	-0.1778 (0.3196)	0.1045 (0.2080)	0.3658 (0.7495)
8	B: Mag7 returns on Flow(t-1)	2022-2025	48	-0.1803 (0.3423)	-0.0050 (0.2812)	0.7257 (0.9134)
9	B: Mag7 returns on Flow(t-1)	2023-2025	36	-0.4550 (0.4251)	-0.0875 (0.2265)	-0.8263 (1.0422)

Out[6]:

	Sample window	N	H2: beta on ASVI8(t-1)	H2 R2	H2 DW	Feedback: beta on Mag7(t-1)	Feedback R2	Feedback DW
0	2016-2025	120	0.0231** (0.0106)	0.287	2.04	2.3110*** (0.8857)	0.092	0.51
1	2020-2025	72	0.0242* (0.0124)	0.330	2.09	2.8419** (1.1391)	0.142	0.50
2	2021-2025	60	0.0267** (0.0131)	0.376	2.04	3.7016** (1.4713)	0.174	0.57
3	2022-2025	48	0.0330** (0.0139)	0.459	2.08	4.5415*** (1.4466)	0.296	0.71
4	2023-2025	36	0.0140 (0.0158)	0.265	1.79	4.4123* (2.6604)	0.206	0.41

In [7]:

```
# Numbers in Section 4.3
m = h1[("2016-2025", "spy")]
print(f"H1 SPY: beta {m.params['ASVI8_lag']:.4f}, t {m.tvalues['ASVI8_lag']:.2f}, p
{m.pvalues['ASVI8_lag']:.3f}")
for f in funds:
    print(f"H1 {f.upper()}: positive in", sum(h1[(w, f)].params["ASVI8_lag"] > 0 for w in
windows), "of 5 windows")
print("H1: significant at 10% in", sum(m.pvalues["ASVI8_lag"] < 0.10 for m in h1.values()),
"of 15 regressions")

print()
for w in windows:
    print(f"{w}:  H1b SPY p {h1b[(w, 'spy')].pvalues['flow_spy_lag']:.3f}  |  "
          f"H2 beta {h2[w].params['ASVI8_lag']:.4f}, p {h2[w].pvalues['ASVI8_lag']:.3f}  |  "
          f"Feedback p {fb[w].pvalues['mag7_ret_lag']:.3f}")
```

```
H1 SPY: beta -0.0053, t -1.32, p 0.185
H1 SPY: positive in 0 of 5 windows
H1 RSP: positive in 1 of 5 windows
H1 VOO: positive in 3 of 5 windows
H1: significant at 10% in 0 of 15 regressions

2016-2025:  H1b SPY p 0.214  |  H2 beta 0.0231, p 0.029  |  Feedback p 0.009
2020-2025:  H1b SPY p 0.577  |  H2 beta 0.0242, p 0.051  |  Feedback p 0.013
2021-2025:  H1b SPY p 0.578  |  H2 beta 0.0267, p 0.041  |  Feedback p 0.012
2022-2025:  H1b SPY p 0.598  |  H2 beta 0.0330, p 0.018  |  Feedback p 0.002
2023-2025:  H1b SPY p 0.284  |  H2 beta 0.0140, p 0.378  |  Feedback p 0.097
```